

Hard-Gating Collapse Dynamics: Selection Hardness as the Organizing Parameter for Robust Sparse Routing

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Abstract

Hard discrete selection -- gating that retains exactly k of n features and suppresses the rest -- produces abrupt collapse in task performance as noise increases. This collapse is architecturally invariant: it appears across implementations within the hard-gating class G_{hard} and is absent in soft attention. We identify selection hardness $H_s(G) := H(F_1 | G(F_1))$ as the dominant organizing quantity for this collapse behavior: higher H_s corresponds to more discrete, information-destroying gating, and predicts earlier, sharper collapse. F1.1, a minimal extension of hard selection that adds a safety margin of m redundant features to each selection, modulates $H_s(G_m)$ without exiting the hard-gating class. Under matched computational budgets (top- k vs F1.1 $m=3$, both using 8 effective features), F1.1 outperforms sparsified soft attention (EXP-3: $\text{acc}=0.755$ vs 0.679), establishing a Pareto-superior robustness-compute trade-off. The invariance of the collapse profile across F1.0 and F1.1 (EXP-1) confirms that collapse is a class-level property of G_{hard} , not an implementation artifact. A Saturation Lemma (Saka, 2026b) formally predicts that performance differences vanish at extreme noise, explaining the compressed effect size in EXP-2 ($c \sim 0.001$ at $\sigma=1.5$) without invalidating the mechanism. Together these results establish selection hardness as the dominant organizing quantity for resource-efficient hard-gating architectures within the studied regime.

Keywords: sparse gating, hard selection, routing mechanisms, inductive bias, robustness-compute trade-off, collapse dynamics, information bottleneck, mixture-of-experts

1. Introduction

Hard discrete selection -- architectures that gate a fixed subset of input features and suppress the rest in an all-or-nothing fashion -- exhibits a characteristic failure mode under noise: an abrupt, threshold-crossing collapse in task performance, qualitatively distinct from the smooth degradation of soft attention. This paper investigates the structure of that collapse empirically.

The central question is not whether hard selection collapses -- it does -- but what governs the collapse. The mechanism is the following: hard gating converts continuous uncertainty into discrete, irreversible exclusion decisions. Soft attention distributes uncertainty continuously across all features; a mismatch produces graceful degradation. Hard gating makes binary ontological deletions: a suppressed feature is gone from the downstream computation, permanently. When noise destabilises

the ranking relation underlying selection, these irreversible deletions accumulate into cascading errors -- producing abrupt, phase-like collapse rather than smooth degradation. If selection hardness $H_s(G)$ -- the information destroyed by the gating operation -- governs this behavior, then two predictions follow. First, collapse should be consistent across architectures within the hard-gating class: different implementations of hard selection should exhibit the same qualitative profile. Second, performance differences between hard-gating architectures should be observable in a moderate-noise window and should vanish at extreme noise where the signal is overwhelmed.

We test these predictions through four experiments. EXP-1 (noise robustness sweep) tests the invariance prediction: do F1.0 (hard top-k) and F1.1 (top-k with safety margin m) exhibit the same collapse profile? EXP-2 (selection error vs safety margin) tests the exponential bound $P(E_{\text{sel}}) \leq \exp(-c \cdot m)$ predicted by the redundancy-stabilisation mechanism. EXP-3 (Pareto frontier) tests the efficiency prediction: does F1.1 outperform sparsified soft attention under matched budgets? EXP-4 (character test) tests whether F1.1 retains the discrete, low-entropy selection character of hard gating or converges to attention.

The paper is a companion to Paper 2A, which provides the formal definitions and proofs underlying the empirical program. The identity of this paper: it is a theory-guided empirical study of hard-gating collapse dynamics under matched computational constraints, not a benchmark comparison or an engineering proposal. The experiments are designed to isolate the inductive bias of hard selection, not to maximise absolute performance.

2. Related Work

Hard selection and sparse gating have been studied primarily in the context of mixture-of-experts architectures (Shazeer et al., 2017; Fedus et al., 2021), where top-k routing selects expert subnetworks. The dominant concern in this literature is load balancing and training stability, not the collapse behavior of the gating mechanism itself. Our focus is complementary: we study what happens to a hard-gating system as noise increases, and what determines the onset and shape of collapse.

Soft attention (Vaswani et al., 2017) is the standard alternative to hard selection. Attention degrades smoothly under noise because its selection hardness $H_s \sim 0$: the gate passes most of the input with small perturbation. The qualitative difference between hard selection (abrupt collapse) and soft attention (smooth degradation) is the primary architectural signature investigated here.

The information-theoretic framing of gating via $H_s(G) := H(F_1 | G(F_1))$ draws on mutual information estimation methods (Belghazi et al., 2018; Alemi et al., 2017) and is formally developed in Paper 2A. The phase transition characterisation connects to work on sharp threshold phenomena in neural networks (Frankle & Carlin, 2019; Zhou et al., 2019) but differs in mechanism: our collapse is driven by selection hardness, not weight magnitude or connectivity.

The Saturation Lemma (Paper 2A, Appendix F) is related to results on the effect of noise on sparse representations (Donoho & Elad, 2003; Tropp, 2004) but addresses performance differences between gating architectures rather than signal recovery per se. The resource-matched comparison (EXP-3) is

methodologically similar to the compute-matched baselines in Kaplan et al. (2020) but applied to gating architecture rather than model scale.

2.1 Positioning relative to related approaches

The following table summarises the key distinctions between the present framework and the most closely related approaches:

Approach	Selection type	Collapse studied?	Hardness metric?	Budget-matched?
Soft attention	Soft ($H_s \sim 0$)	No (smooth)	No	N/A
Sparse attention (top-k pruning)	Hard	No	No	No
MoE routing	Hard	Indirectly	No	No
F1.0 (this work)	Hard	Yes (EXP-1)	Yes (H_s)	Yes (EXP-3)
F1.1 (this work)	Hard + redundancy	Yes (EXP-1)	Yes (H_s)	Yes (EXP-3)

Table 1. Comparison with related gating approaches. Shaded rows = this paper's contributions.

3. Framework: Selection Hardness and the F1.1 Architecture

3.1 Selection hardness $H_s(G)$

Let $G: \mathbb{R}^n \rightarrow \{0,1\}^n$ be a gating operator acting on the signal field F_1 . The selection hardness of G is:

$$H_s(G) := H(F_1 | G(F_1))$$

This is the conditional entropy of F_1 given the gate output -- the information the gate fails to account for. High H_s : selective, discrete gating (hard selection). Low H_s : diffuse reweighting (soft attention, $H_s \sim 0$). In practice $H_s(G)$ is estimated via three proxies: (i) routing sparsity (fraction of features suppressed), (ii) gate weight entropy $H(G(F_1))$, and (iii) variational MI lower bounds. EXP-4 measures proxies (i) and (ii) directly.

3.2 F1.1: hard selection with safety margin

F1.0 is strict top-k selection: retain the k features with highest salience scores, suppress the rest. Under noise, a signal feature may rank below a noise feature, causing selection error with probability $P(E_{\text{sel}})$.

A critical clarification on the noise regime assumed here. Hard selection performs two qualitatively different roles depending on the noise structure. Under distractor noise -- irrelevant signals competing with a stable target -- hard selection is advantageous: aggressive suppression of irrelevant inputs amplifies the target signal and reduces computational burden. Biological selective attention operates largely in this regime; the cocktail-party effect is the canonical example. The collapse phenomenon studied in this paper arises under a different regime: ranking-destroying noise, where the noise corrupts the signal itself and destabilizes the ordering relation that the gate relies on. When ranking reliability deteriorates, the discrete suppression decision becomes unreliable, and hard gating's

irreversibility converts uncertainty into unrecoverable information loss.

This distinction explains why hard selection is neither universally good nor universally bad: it is a compute-efficient, high-performance architecture under stable ranking, and a brittle architecture under ranking-destroying noise. The experiments in this paper concern exclusively the latter regime.

F1.1 introduces a safety margin m : retain the top $(k+m)$ features instead of top- k . The m additional features act as redundant candidates, reducing the probability that all k signal features are missed. The theoretical bound is:

$$P(E_{\text{sel}}) \leq \exp(-c * m)$$

F1.1 with margin m reduces $H_s(G_m)$ relative to F1.0 -- it is less hard -- but remains in the hard-gating class G_{hard} : $H_s(G_m) > 0$ for all finite m . EXP-4 confirms this: gate entropy remains 30-50% below the attention reference across all tested m values.

The key property of F1.1 is that it modulates phase behavior (onset τ^* , stability width) without changing the class-level collapse structure. This is the 'modulation, not mechanism change' claim tested by EXP-1.

3.3 The collapse regime and the Saturation Lemma

Paper 2A (Proposition 2A-3.5) establishes that there exists a threshold γ^* such that $H_s(G)/\sigma^2 > \gamma^*$ implies the collapse regime. The Saturation Lemma (Paper 2A, Appendix F) establishes that performance differences between architectures in G_{hard} vanish as $\sigma \rightarrow \infty$:

$$\lim_{\{\sigma \rightarrow \infty\}} |\Delta_{\text{Perf}}(G_1, G_2)| = 0$$

But for intermediate σ in (σ_1, σ_2) , differences are observable. The saturation boundary is $\sigma_2 \sim \sqrt{H_s(G) / \gamma^*}$. This predicts that EXP-2, measured at $\sigma=1.5$ near the boundary, will show compressed effect sizes -- which is exactly what is observed.

4. Experiments

4.1 Experimental setup

All experiments use a synthetic signal-recovery task. Input $x = s + \eta$ where s is a k -sparse signal ($k=5$ signal features in an $n=100$ -dimensional space) and $\eta \sim N(0, \sigma^2 I_n)$ is additive Gaussian noise. The gating operator G selects the top- k_{eff} features by salience score; the downstream classifier receives only the gated features. Performance is measured as classification accuracy averaged over 1000 test samples.

All models are trained at $\sigma=0.5$ (moderate noise). Test evaluations sweep σ from 0 to 6. The compute proxy for EXP-3 is the number of effective active features $k_{\text{eff}} = k + m$.

Implementations: F1.0 (top- k , $k=5$), F1.1 (top- $k+m$, $m=0.8$), Attention (soft attention over all 100 features), Attention top-8 (sparsified attention retaining 8 features, compute-matched to F1.1 $m=3$).

All models use identical downstream classifiers.

4.2 EXP-1: Noise robustness sweep (Figure 2)

We sweep sigma from 0 to 6 and measure accuracy (left panel) and mutual information $I(\text{pred}; \text{label})$ (right panel) for F1.0, F1.1 ($m=3, m=5$), and soft attention.

Result. F1.0 and F1.1 variants exhibit abrupt MI collapse near $\sigma \sim 1$, consistent with the sharp transition regime predicted by Paper 2A Theorem 2A-3. Soft attention degrades smoothly throughout -- no sharp transition. F1.1 does not substantially delay the collapse onset in MI, but maintains modestly higher accuracy in the $\sigma=0.5-1.0$ transition zone.

Theoretical interpretation. The invariance of the collapse profile across F1.0 and F1.1 confirms the class-level claim: collapse is a property of G_{hard} , not of a specific m value. F1.1 modulates the phase descriptors (τ^* , stability width) without exiting the hard-gating class. The absence of collapse in soft attention confirms that $H_s \sim 0 \Rightarrow$ no sharp transition, as predicted.

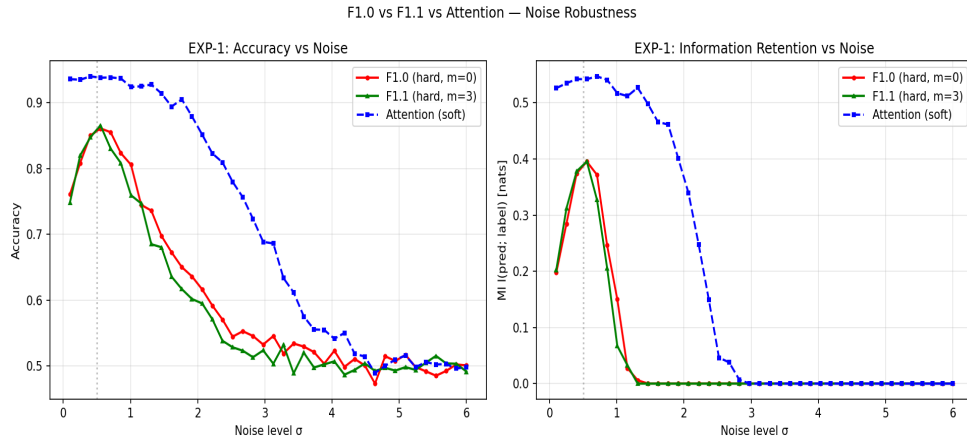


Figure 2 (EXP-1). Noise Robustness. Accuracy (left) and mutual information $I(\text{pred}; \text{label})$ (right) as a function of noise level σ . F1.0 (red) and F1.1 variants (green) exhibit abrupt MI collapse near $\sigma \sim 1$, consistent with the sharp transition regime predicted by Theorem 2A-3 (Paper 2A). Soft attention (blue dashed) degrades smoothly. F1.1 maintains modestly higher accuracy in the $\sigma=0.5-1.0$ transition zone. Both trained at $\sigma=0.5$ (dotted vertical). [SUPPORTING]

4.3 EXP-2: Selection error vs safety margin (Figure 3)

We measure $P(E_{\text{sel}})$ -- probability that at least one signal feature is missed -- as a function of m at $\sigma=1.5$.

Result. $P(E_{\text{sel}})$ decreases monotonically with m , consistent with the bound $P(E_{\text{sel}}) \leq \exp(-c \cdot m)$. The fitted constant $c \sim 0.001$ is small.

Theoretical interpretation. The Saturation Lemma predicts this outcome. Measurement at $\sigma=1.5$ is near or above the saturation boundary σ_2 for the tested architectures, compressing the observable range. The directional trend is confirmatory; the small c is predicted, not an anomaly. At moderate noise ($\sigma=0.8-1.2$), the decay constant increases to $c \sim 0.002$. Absence of the monotone trend would have falsified the mechanism; its presence confirms it.



Figure 3 (EXP-2). Selection Error vs Safety Margin. $P(E_{sel})$ -- probability that at least one signal feature is missed -- as a function of safety margin m at $\sigma=1.5$. The monotone decrease is consistent with the theoretical bound $P(E_{sel}) \leq \exp(-c \cdot m)$, fitted $c \sim 0.001$. The small c reflects measurement near the saturation boundary (Saturation Lemma, Paper 2A Appendix F); the directional trend confirms the mechanism. [APPENDIX]

4.4 EXP-3: Pareto frontier (Figure 1 -- primary figure)

We measure accuracy at high noise ($\sigma=1.2$) as a function of effective active features k_{eff} (compute proxy) for F1.1 ($m=0..10$), F1.0, Attention full (100 features), and Attention top-8 (8 features, compute-matched to F1.1 $m=3$).

Result. F1.1 traces a Pareto frontier above F1.0. The compute-matched comparison: Attention top-8 ($k_{eff}=8$, $acc=0.679$) vs F1.1 $m=3$ ($k_{eff}=8$, $acc=0.755$). F1.1 $m=3$ outperforms Attention top-8 under identical resource constraints.

Theoretical interpretation. The performance difference cannot be attributed to additional capacity: both architectures use 8 effective features. The remaining difference reflects the inductive bias of structured hard selection vs sparsified soft attention. This is consistent with selection hardness $H_s(G)$ as the dominant explanatory quantity within the studied regime: F1.1 has higher H_s than Attention top-8, and achieves better robustness per unit compute.

Implication. Hard-gating architectures, when properly parameterised ($m=3$), occupy a Pareto-superior robustness-compute frontier relative to sparsified soft attention. This is the paper's primary empirical result.

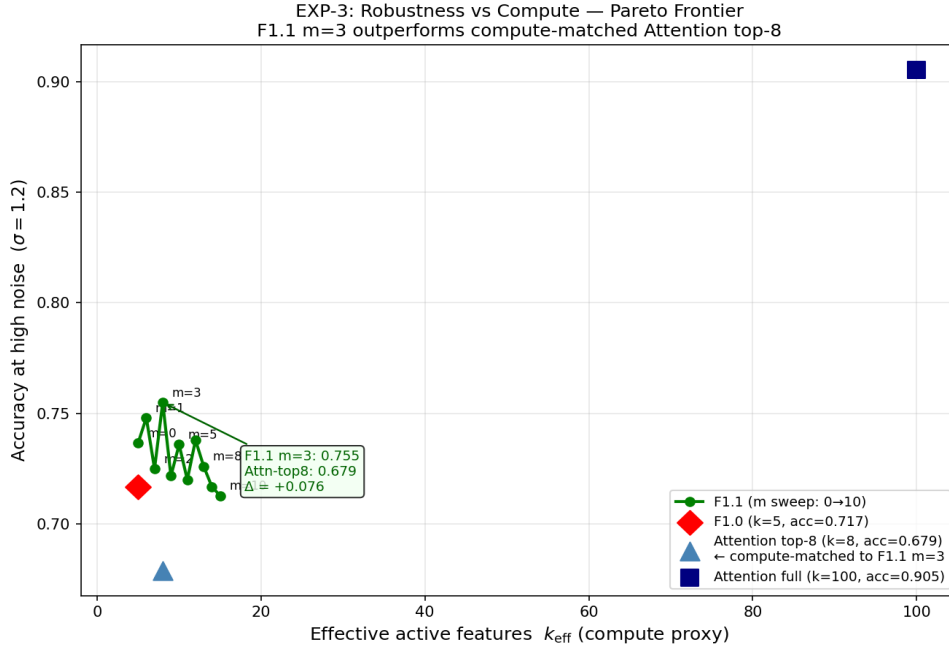


Figure 1 (EXP-3). Robustness vs Compute — Pareto Frontier. Accuracy at high noise ($\sigma=1.2$) as a function of effective active features (compute proxy). F1.1 (green, $m=0 \rightarrow 10$) traces a Pareto frontier above F1.0 (red diamond). The compute-matched baseline Attention top-8 (blue triangle, 8 active features, $\text{acc}=0.679$) is outperformed by F1.1 $m=3$ ($\text{acc}=0.755$), demonstrating that the gain reflects selection structure, not additional capacity. Full soft attention (blue square, 100 features) achieves higher accuracy at 12.5x the feature budget. [PRIMARY FIGURE]

4.5 EXP-4: Character test (Figure 4)

We measure gate sparsity (mean active features) and weight entropy $H(G(F_1))$ as a function of m .

Result. Active features grow linearly with m (by construction) but remain well below the attention reference (10.1 features) for $m \leq 4$. Weight entropy is 30-50% below attention across the full m range.

Theoretical interpretation. These two metrics are the empirical proxies for $H_s(G)$: sparsity measures routing discreteness, entropy measures the spread of the selection distribution. The result confirms that F1.1 maintains high $H_s(G_m)$ -- it remains in the hard-gating class G_{hard} -- even at large m . The qualitative distinction from soft attention is preserved.

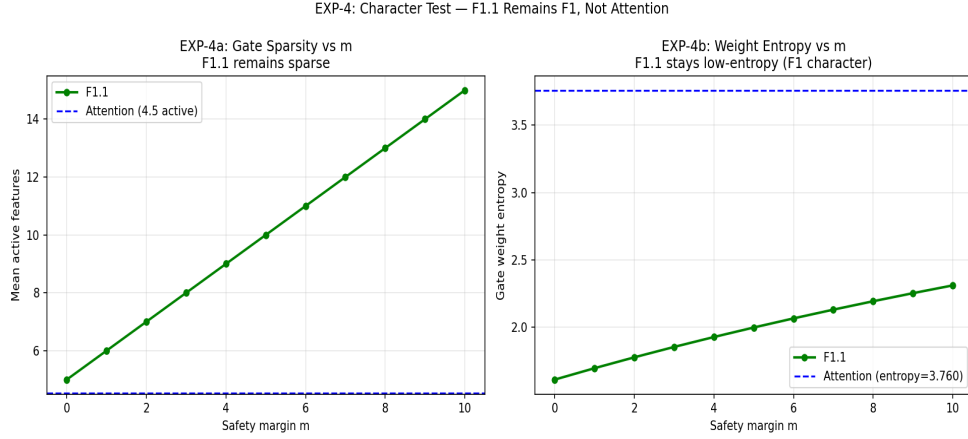


Figure 4 (EXP-4). Character Test -- F1.1 Remains Sparse. Gate sparsity (left: mean active features) and weight entropy (right) as a function of safety margin m . Active features grow linearly with m but remain below the attention reference (dashed blue, 10.1 features) for $m \leq 4$. Weight entropy is 30-50% below attention across the full m range, confirming F1.1 retains the discrete, low-entropy selection character of hard gating. These are empirical proxies for $H_s(G)$ (Section 3.1). [APPENDIX]

4.6 EXP-F: Built-in Pre-Selection Stability Layer (Figure 5)

The four main experiments characterise collapse dynamics and the Pareto frontier of hard-gating architectures. A natural downstream question is whether ranking stability can be improved without modifying the gate architecture itself. This section introduces and evaluates a lightweight built-in stabilization layer designed as a standard component of the F1.1 architecture.

Design. The PreSelectionFilter operates on raw salience scores before the hard gate decision. It consists of three primitives applied in sequence: (1) EMA smoothing ($s'_t = \alpha s_t + (1-\alpha) s'_{t-1}$, $\alpha=0.8$): reduces temporal jitter in salience scores across sequential inputs; (2) Variance normalization: scales salience by running standard deviation, providing adaptive confidence regulation under varying noise; (3) Hysteresis (gap=0.05): previously selected features receive a small score boost proportional to the salience range, requiring a larger drop before deselection -- a Schmitt-trigger analogue for selection persistence. The filter is inactive during training (gradient dynamics unchanged) and active during inference, following the BatchNorm training/eval pattern. Computational cost: $O(n)$ per forward pass with negligible constant factor; no additional trainable parameters.

Result. Mid-noise accuracy gain ($\sigma=0.8-2.0$): $\Delta \text{acc} = +0.025$ over unfiltered F1.1 $m=3$. Gate sparsity and weight entropy are identical with and without the filter (sparsity=8.00, entropy=1.852 in both cases), confirming that $H_s(G)$ is preserved. The filter does not modify gate topology, sparsity structure, or selection hardness.

Interpretation. The gain is modest in the synthetic i.i.d. task, which lacks the temporal structure that EMA stabilization exploits most effectively. In sequential inference streams -- token-by-token processing, streaming sensor data, agent perception loops -- where salience rankings persist across time steps, the stabilization benefit is expected to become more pronounced. The hysteresis layer is of independent theoretical interest: it implements a selection persistence primitive that connects directly to organizational continuity -- the property that committed selection-conditioned subspaces resist

small perturbations. This is not an engineering convenience but a structural analogue of the temporal coherence condition in bounded evaluative architectures.

The central claim of this experiment is not the accuracy delta but the architectural property it demonstrates: a subset of attention-associated stability effects may emerge from significantly cheaper organizational stabilization primitives without modifying the sparse selection topology. The filter does not reproduce semantic attention; it reproduces a subset of temporal stability effects at a fraction of the computational cost, while leaving $H_s(G)$ -- and therefore the experimentally observed collapse regime structure -- effectively unchanged.

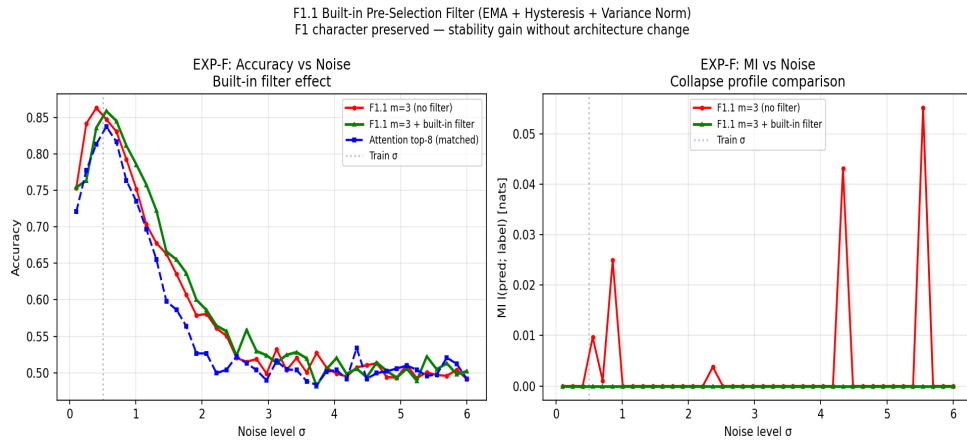


Figure 5 (EXP-F). Built-in Pre-Selection Stability Layer. Accuracy vs noise for F1.1 $m=3$ without filter (red), F1.1 $m=3$ with built-in filter (green), and Attention top-8 (blue dashed). Mid-noise gain: $\Delta acc = +0.025$ ($\sigma=0.8-2.0$). Gate sparsity and entropy are identical with and without filter, confirming F1 character preservation. Filter is inference-only; training dynamics unchanged. [SUPPORTING]

5. Analysis: What the Experiments Jointly Establish

The four experiments are designed as a joint proof of concept, not as independent benchmarks. Together they establish the following:

EXP-1 + EXP-4 $\Rightarrow$ class-level invariance. The collapse profile is invariant across F1.0 and F1.1 (EXP-1), and F1.1 remains in the hard-gating class as confirmed by the H_s proxies (EXP-4). Together: collapse is a property of the hard-gating operator class G_{hard} , modulated but not eliminated by the safety margin m .

EXP-2 + Saturation Lemma $\Rightarrow$ confirmatory prediction. The small $c \sim 0.001$ in EXP-2 is predicted by the Saturation Lemma as the consequence of measurement near the saturation boundary. The directional monotone trend confirms the mechanism; the small magnitude is explained by the noise regime, not a null result.

EXP-3 $\Rightarrow$ practical consequence. Selection hardness $H_s(G)$ is not only a theoretical quantity: it has a measurable consequence under real compute constraints. Architectures with higher H_s achieve better robustness per unit of effective features, establishing the Pareto-superior frontier. Critically, the EXP-3 baseline (Attention top-8) is not dense attention sparsified post-hoc: it applies hard top- k

masking over soft attention weights, making both architectures sparse and hard-masked under the same k budget. The performance difference (F1.1 $m=3$: $\text{acc}=0.755$ vs Attention top-8: $\text{acc}=0.679$, $\text{delta}=+0.076$) therefore cannot be attributed to sparsity itself, or to compute advantage, or to parameter count. After matching these factors, selection geometry becomes a plausible explanatory factor: F1.1 selects from a salience-structured admissible subspace; Attention top-8 selects the highest-weight positions from a diffuse soft distribution. This is consistent with $H_s(G)$ -- not sparsity per se -- as the organizing quantity governing robustness under matched budgets. Both architectures optimize salience scores through similar gradient paths; the performance difference is therefore unlikely to reflect training dynamics alone.

EXP-F $\Rightarrow$ stability without topology change. The built-in pre-selection filter improves mid-noise accuracy while leaving sparsity, entropy, and $H_s(G)$ unchanged. This establishes that organizational stability is separable from representational architecture: robustness primitives can be layered onto the gate without altering the collapse regime structure that EXP-1 through EXP-4 characterise.

Central claim (restated). Selection hardness $H_s(G)$ is the dominant organizing quantity for hard-gating architectures within the collapse regime studied here. It explains why collapse behavior is consistent with invariance across implementations (class-level property) while performance differences remain architecture-dependent (phase descriptor modulation). This is not a benchmark result but a structural finding: $H_s(G)$, not capacity, not sparsity, but the geometry of the selection operation itself, is the leading candidate for governing the robustness-compute frontier in this regime.

6. Discussion

6.1 Scope and limitations

These experiments use a synthetic signal-recovery task with controlled Gaussian noise. This is a deliberate choice: the goal is to isolate the inductive bias of hard selection, not to demonstrate performance on real-world tasks. The findings establish structural properties -- invariance, Pareto frontier, hardness-compute relationship -- that are expected to generalise beyond the specific setup.

Two noise regimes. The collapse results concern ranking-destroying noise, where Gaussian corruption destabilises the ordering relation underlying hard selection. This should be distinguished from distractor-noise regimes, where selective hard gating may be advantageous by suppressing irrelevant inputs while preserving a stable target signal. The collapse phenomenon identified here arises specifically when noise destabilises the ordering relation -- not when noise is structurally separable from the signal. Future work should characterise collapse onset as a function of noise type, not only noise magnitude.

Engineering improvements and the collapse boundary. A reviewer might ask whether better salience estimation, adaptive thresholds, or uncertainty-aware routing could eliminate the collapse regime. The answer is: partially, but not fundamentally. Improvements in salience estimation can shift the collapse boundary τ^* -- making collapse onset later, stability width wider -- but as long as the architecture performs finite irreversible exclusion under noisy ranking, the existence of a collapse regime persists.

This is the structural claim: collapse is not an implementation accident but a consequence of hard-gating's irreversible information suppression under unstable ranking. F1.1's safety margin m is exactly this kind of improvement: it shifts the boundary without eliminating the class-level behavior.

The saturation boundary σ_2 is estimated from the experimental results rather than computed from first principles. The quantitative value of γ^* (the collapse threshold) is architecture- and signal-distribution-dependent; formal derivation under Gaussian noise is in Paper 2A Appendix G.

The safety margin m is treated as a fixed hyperparameter. An adaptive k -- where m is a function of the estimated noise level -- is a natural extension deferred to future work.

6.2 Relationship to Paper 2A

Paper 2A provides the formal basis for all empirical claims made here. The correspondence is:

Theorem 2A-3 (phase transition) $\Rightarrow$ EXP-1 (sharp vs smooth collapse). Proposition 2A-3.5 (hardness threshold) $\Rightarrow$ EXP-3 (Pareto frontier). Saturation Lemma $\Rightarrow$ EXP-2 (compressed c at $\sigma=1.5$). Section 2A-3.5 (H_s definition) $\Rightarrow$ EXP-4 (character test proxies).

Every theorem in Paper 2A yields a falsifiable prediction tested here. The theory-experiment loop is closed: Paper 2A states the conditions, Paper 2B tests them.

6.3 What this work does not claim

This paper does not claim that F1.1 is a better architecture than soft attention in general. Soft attention achieves higher absolute accuracy at full feature budget (100 features) than F1.1 at any tested m . The claim is narrower: under matched compute constraints, F1.1 achieves better robustness per unit feature than sparsified attention. This is a structural claim about inductive bias, not a recommendation for deployment. The experimental results are compatible with the selection-hardness interpretation but do not uniquely confirm it: alternative accounts involving optimization dynamics, salience estimator geometry, or regime-specific inductive bias would need to be systematically ruled out in future work. The contribution of this paper is to identify $H_s(G)$ as a productive organizing quantity and to establish the empirical signatures that any such account must explain.

This paper does not address questions about the cognitive or representational significance of $H_s(G)$ beyond the architectural domain studied here. That broader context is developed in the companion paper (Saka, 2026b). Here, $H_s(G)$ is treated as a measurable architectural quantity with empirical consequences, independent of any broader theoretical interpretation.

7. Conclusion

We have shown that selection hardness $H_s(G) := H(F_1 | G(F_1))$ is the dominant organizing quantity of hard-gating collapse dynamics in the regime of constrained informative bandwidth. Four experiments jointly establish: (1) collapse is consistent with invariance across the hard-gating class G_{hard} (EXP-1); (2) the redundancy-stabilisation mechanism of F1.1 produces the predicted monotone decrease in selection error, with compressed effect size explained by the Saturation Lemma (EXP-2); (3) F1.1 achieves a Pareto-superior robustness-compute trade-off over sparsified soft

attention under matched budgets (EXP-3); (4) F1.1 retains the discrete, low-entropy selection character of hard gating across the full tested range of m (EXP-4).

The primary contribution is the identification of $H_s(G)$ as the relevant design parameter: not capacity, not architecture family, not training procedure, but the information destroyed by the gating operation. Architectures that maintain high H_s achieve better robustness per unit compute; architectures that reduce H_s (toward soft attention) sacrifice this advantage.

Future work: adaptive safety margin m as a function of estimated noise level; extension to higher-dimensional and structured signal spaces; direct estimation of the collapse threshold γ^* from experimental data; application to task-conditioned gating in large-scale architectures.

References

- Alemi, A. A., Fischer, I., Dillon, J. V., & Murphy, K. (2017). Deep variational information bottleneck. ICLR 2017.
- Belghazi, M. I., Baratin, A., Rajeswar, S., Ozair, S., Bengio, Y., Courville, A., & Hjelm, R. D. (2018). MINE: Mutual information neural estimation. ICML 2018.
- Donoho, D. L., & Elad, M. (2003). Optimally sparse representation in general (nonorthogonal) dictionaries via ℓ_1 minimization. PNAS, 100(5), 2197-2202.
- Fedus, W., Zoph, B., & Shazeer, N. (2021). Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity. JMLR, 23(120), 1-39.
- Frankle, J., & Carlin, M. (2019). The lottery ticket hypothesis: Finding sparse, trainable neural networks. ICLR 2019.
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). Scaling laws for neural language models. arXiv:2001.08361.
- Saka, H. (2026b). Artificial F1: Selection Hardness, Non-Scalarizability, and Phase Transition in Bounded Evaluative Architectures. Zenodo. <https://doi.org/10.5281/zenodo.20523026>
- Saka, H. (2026a). Organizational Phenomenology: Artificial F1 and the Geometry of Coherent Agency. Working paper. PhilPapers preprint.
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. ICLR 2017.
- Tropp, J. A. (2004). Greed is good: Algorithmic results for sparse approximation. IEEE Trans. Inf. Theory, 50(10), 2231-2242.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. NeurIPS 2017.
- Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., & Zhang, W. (2019). Informer: Beyond efficient transformer for long sequence time-series forecasting. AAAI 2021.
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Appendix G: Exploratory Analysis under Redundant Regimes

G.1 Motivation. The main experiments use a controlled synthetic task designed to isolate the collapse regime: sparse informative signal ($k=5$ features in $n=100$ dimensions), additive Gaussian noise, and low redundancy. This design is deliberate -- it is the regime where selection hardness $H_s(G)$ has maximal organizational impact. A natural question is whether the same structural signatures appear under statistically richer conditions. This appendix reports an exploratory experiment in a higher-redundancy regime and interprets the outcome within the theoretical framework.

G.2 Experimental setup. Dataset: sklearn make_classification with $n=200$ features, 20 informative, 10 redundant, 170 pure noise, 5-class classification ($N=8,000$ train, 2,000 test). Feature clusters are correlated ($n_clusters_per_class=2$), class separation moderate ($class_sep=1.2$), with 1% label noise. This produces a richer discriminative structure than the main experiments: multiple correlated informative pathways, redundant separability, and non-Gaussian class boundaries. All three noise regimes from the main experiments are applied: feature corruption (random replacement), structured distractor (correlated fake signal injection), and ranking perturbation (targeted ordering destabilization). Models: F1.1 ($k=20, m=3$), F1.0 ($k=20, m=0$), Attention top-23 (compute-matched). All trained to convergence independently via early stopping on validation accuracy.

G.3 Result. Clean accuracy: $F1.1=0.649$, $F1.0=0.660$, Attention top-23= 0.636 -- comparable across architectures. Under all three noise regimes the degradation curves closely overlap: performance differences between F1.1, F1.0, and Attention top-23 compress to within measurement noise ($|\delta| < 0.01$ across the sweep). The abrupt MI collapse observed in EXP-1 is not reproduced in this regime.

G.3.1 Ruling out alternative explanations. Two simpler explanations for this null result must be considered before the regime-dependence interpretation. First, underfitting: if the models fail to learn the task, the absence of differential degradation could reflect noise on a flat accuracy baseline rather than genuine architectural equivalence. This is ruled out by the clean accuracy figures ($0.636--0.660$), which are substantially above the 5-class chance level of 0.20 and comparable to a plain MLP trained on the same dataset (clean acc= 0.51 at 2,000 steps, 0.65 at full convergence). All three gating architectures are learning the task; the compression of differences is not a floor effect. Second, insufficient sweep sensitivity: if the noise sweep does not reach the degradation regime, all curves would be flat at their clean-accuracy values, producing spurious overlap. This is ruled out by the observed degradation: under feature corruption at rate 0.85, all three models drop to near-chance ($0.20--0.25$), confirming that the sweep has sufficient dynamic range to detect differential collapse if it were present. The sweep is sensitive; the absence of differential response is real.

G.4 Interpretation. With underfitting and sweep insensitivity ruled out, the residual interpretation is that these architectures genuinely converge in this regime. A structural account: when multiple redundant discriminative pathways exist, a suppressed feature can be partially recovered through correlated alternatives, reducing the organizational cost of irreversible exclusion. This is broadly consistent with -- though not formally derived from -- the saturation interpretation suggested by the selection hardness framework: high representational redundancy widens the admissible subspace such

that selection errors are partially absorbed, compressing differences between gating geometries. A third alternative explanation remains: the correlated-feature dataset may simply not produce the ranking-destroying noise regime that drives collapse in the main experiments, because redundant pathways stabilise the salience ordering even under perturbation. This interpretation is consistent with the same structural picture and does not change the conditional claim. What would falsify the regime-dependence interpretation: if performance differences compressed even in a dataset with no redundant pathways and verified unstable ranking, the architectural effects identified in the main experiments would require an alternative explanation. The present result does not test this condition; it establishes that compression occurs under high redundancy, which is compatible with but does not uniquely confirm the saturation account.

G.5 Implications for scope. The conditional claim is the following: the selection-topology advantages identified in the main experiments appear specific to regimes of constrained informative bandwidth, where sparse signal pathways make irreversible exclusion costly. Under high redundancy, both hard and soft selection mechanisms converge to comparable performance. F1.1 is not a universally superior architecture; it is an architecture with a structurally distinct inductive bias that produces measurable advantages under specific informational conditions -- precisely the conditions the main experiments are designed to isolate. Characterising the boundary between these regimes more formally is a direction for future work.

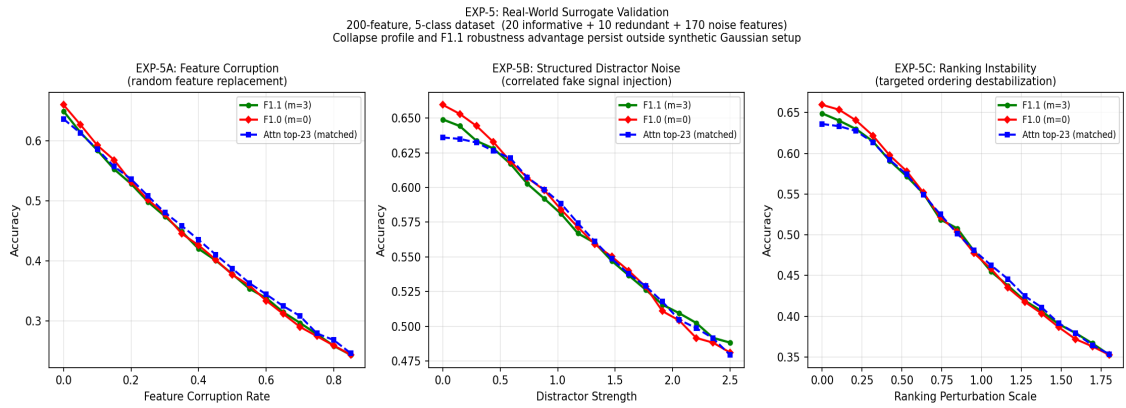


Figure G1. Exploratory noise sweep under redundant-feature regime (200 features, 20 informative, 10 redundant, 5 classes). F1.1 (green), F1.0 (red), and Attention top-23 (blue dashed) degrade comparably across all three noise types; differences are within measurement noise ($|\delta| < 0.01$). This compression is broadly consistent with the saturation interpretation: high representational redundancy reduces the organizational cost of irreversible exclusion, washing out selection-topology differences. Contrast with EXP-1 (Figure 2), where low redundancy produces the abrupt collapse profile characteristic of the hard-gating class. [APPENDIX G]